Automation in Construction 138 (2022) 104257

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Digital twin: Stability analysis for tower crane hoisting safety with a scale model

Weiguang Jiang [a,b], Lieyun Ding [a,b,*], Cheng Zhou [a,b]

[a] School of Civil & Hydraulic Engineering, Huazhong University of Science & Technology, Wuhan, Hubei, China
[b] Hubei Engineering Research Center for Virtual, Safe and Automated Construction(ViSAC), Wuhan, Hubei, China

ARTICLE INFO

Keywords:
Digital twin
Scale model
Unsafe hoisting behaviors
Tower crane stability

ABSTRACT

Unsafe hoisting behaviors are the fuse of tower crane overturning safety accident. However, the hoisting operation has a certain degree of randomness and invisibility, and it is extremely dangerous to carry out the experiment on the construction site, which makes the research on the relevant regularity very scarce. To simulate different hoisting behaviors as realistically as possible and analyze their dynamic effects on the tower crane, a digital twin (DT) framework of human–machine interaction construction activities is constructed based on the scale model in this study. Through experimental study, the stability change law of the tower crane with the incremental dynamic load is obtained based on the collapse probability level function. Experiment results show that tilt hoisting is the most dangerous behavior causing the tower crane instability. Tower crane foundation and masts are the most vulnerable components to be disturbed by unsafe hoisting behaviors, which can be used as key areas for sensor deployment. This paper can support the detection of the dangerous parts of the tower crane structure, and the monitoring system design for unsafe hoisting behaviors will be carried out in the future.

1. Introduction

The tower crane is a high-rise structure, and its collapse risk is extremely significant caused by hoisting error (e.g., tilt hoisting, sudden braking, and sudden unloading). Wiethorn conducted a detailed analysis of 701 lifting and hoisting accidents that occurred in the past 30 years, 92.9% of all crane accidents had some type of human intervention. Among them, the accident responsibilities of the tower crane driver and the rigger both accounted for more than 20% [1]. In South Korea, Hong Kong China and other regions, researchers have found that workers who do not comply with the operation procedures are the main cause of tower crane accidents [2–4].

Analyzing the dynamic effects of tower cranes under unsafe hoisting behaviors is a prerequisite for on-site safety management, which means that managers can focus more on the weakest structural parts and deploy sensors at the most sensitive locations. However, these unsafe hoisting behaviors and habits are especially tricky to detect because of their indistinct feature in the actual construction. The hoisting excitation effects on the stability of the machinery itself also cannot be comprehensively analyzed. At present, a mainstream solution is to use the simulation method, where the external excitation is applied to the model as a sequential load by expertise [5]. However, the simulation results always have difficulty reflecting the real operation status accurately because of too much boundary hypothesis condition and the randomness of driving behaviors. Another way is to design a scale model in which to test the real running state. Some studies have proposed the hoisting sway control method of tower cranes by the scale model [6]. In the field of manufacturing, the experimentable DT based on the scale model have been established to obtain the small sample data and conduct cross-over analysis of systems in different fields and extreme condition simulation (e.g., working on mars) [7,8]. The experimental method of the scale model DT provides a way to analyze the safety risk problem of human-machine interaction construction, such as hoisting.

For the construction process, DT has produced a lot of practical value. Boje's research suggested that DT can play an important role in construction site monitoring, quality control, safety management, scheduling, and visual communication [9]. Meanwhile, the establishment of a construction DT is even more challenging because most construction activities are the process of human-machine interaction. Among them, workers' control, command behaviors, and habits will greatly affect the safety state of machinery, and the vibration and noise generated by machinery will also affect workers' emotions and health

* Corresponding author at: School of Civil & Hydraulic Engineering, Huazhong University of Science & Technology, Wuhan, Hubei, China.
E-mail address: dly@hust.edu.cn (L. Ding).

https://doi.org/10.1016/j.autcon.2022.104257
Received 29 April 2021; Received in revised form 14 February 2022; Accepted 6 April 2022
Available online 15 April 2022
0926-5805/© 2022 Elsevier B.V. All rights reserved.

[10]. Nowadays, some studies have attempted to set up a DT for construction workers or machinery. For example, a DT construction site could actively track the personnel trajectory and behavior data for safety risk management [11]. DT-based training and monitoring system was established for the drivers of construction machinery [12]. An increasing number of control fault diagnosis systems of construction machinery was developed by DT [13]. However, none of the above-mentioned studies considers construction activities as a system. In other words, it only represents the independent state of the workers or machinery, without reacting and analyzing the result of their interaction. For hoisting activities, the existing construction DT cannot perform effective analysis.

To obtain the influence law of different hoisting behavior on the stability of tower crane. In this study, A DT of the tower crane is established to provide a virtual testbed for dynamic and stability analysis of the different unsafe hoisting behaviors. Specific innovation work is reflected: (i) propose a DT system framework of human-machine interaction construction activities to analysis hoisting behavior, which is confirmed based on a lean scale model; (ii) expound the load excitation effect of the vertical hoisting, tilt hoisting, sudden braking, and sudden unloading; (iii) the collapse probability level of different tower crane components are established, and the instability evolution process under the unsafe hoisting behavior is described. This study provides a theoretical basis for monitoring the unsafe hoisting behavior of the tower crane.

The remainder of the paper is organized as follows. Section 2 introduces the research status of construction machinery DT, tower crane safety monitoring, modeling, and simulation method. Section 3 presents the DT of the tower crane proposed in this work. Section 4 describes the basic process of hoisting experiments and the results of dynamic analysis. Section 5 comprehensively discusses the global and local stability of the tower crane. Section 6 concludes this study.

2. Related work

2.1. DT for construction machinery

DT for construction machinery is to establish the virtual machinery prototype (e.g., structural, mechanism, and working condition), where real-time operating data will be integrated and analyzed to achieve lifecycle optimal management of production design, manufacturing, operation, and maintenance [14]. In the design phase, DT can greatly reduce the cost of trial and error. As early as 1998, DiMaio proposed the concept of excavator virtualization by using virtual environments to evaluate and improve excavator performance [15]. In addition, DT-driven construction machinery design can integrate expert knowledge, real-time working data, and end-user feedback to maximize design co-optimization [16].

In the manufacturing phase, DT has been used as a management tool for flexible production and customized production. At this point, the mechanical DT also needs context-related elements such as production lines and operators in the factory. For instance, Lim realized the rapid reconfiguration of the tower crane product family (Liebherr) through DT [17]. Thomas used DT for the upgrade of the intelligent product line of excavators [18]. In the operation and maintenance phase, DT provides support for preventive maintenance and fault diagnosis of the construction machinery. Zhidchenko analyzed the hydraulic cylinder pressure data of mobile construction machinery based on DT to estimate engine fatigue life [19,20]. Moi established a DT of the small-scale knuckle boom crane to analyze the partial pressure data for predictive maintenance and product life cycle management [21]. Some research has also embedded the sequential Markov chain Monte Carlo simulation model in DT to optimize the maintenance cycle of construction machinery [22]. In general, DT has become a part of the smart factory, and its research path mainly includes modeling, experiment, analytics, and learning [23]. However, a research gap still needs to be filled, that is, DT for machinery in the construction site.

Compared with DT in the design and manufacturing phase, that in the construction site is more complicated because of the complexity and various uncertain factors. Currently, an increasing number of researchers are studying the data acquisition and analysis method of construction machinery. On the one hand, some of them use machine vision and inertial measurement unit technology to estimate the posture of excavators, bulldozers, and cranes [24,25]. Some mechanical and electrical sensors are used to collect the data of mechanical operation performance. On the other hand, data analysis is a process in which DT generates extended value. Sepasgozar combined AR technology and DT data of excavators, tower cranes, and shield machines for driving training [26]. Deryabina discussed the application of DT in the optimization of the whole mine mining process of the excavator digging, crane hoisting, and loader transportation [27]. However, DT and other information technology applications are scarce for construction safety analysis [28,29], especially for establishing the DT model and conducting the data analysis of human-machine interaction operation on the tower crane.

2.2. Tower crane safety monitoring

Tower cranes are the most common machinery in the construction site, and hoisting activities are also a construction link with frequent safety accidents and extremely high safety risks. Researchers have conducted considerable research on tower crane safety monitoring.

2.2.1. Anti-collision with surrounding objects. The reason for the collision risk is that tower cranes and surrounding obstacles (building structure and workers on site) are mutually exclusive in construction space. Collision accidents can be avoided if the operation attitude of the tower crane and the position of the obstacle can be collected in real time. On the one hand, researchers have used ultra-wide band, ZigBee, visual tracking, and other methods to locate the three motion trajectories of the tower crane, when lifting, rotation, and luffing [30–33]. The distance threshold is used to avoid obstacles in different operations of the tower crane. On the other hand, ultrasonic sensors, 3D scanners, and multi-sensor fusion methods have been used to rapidly reconstruct the working space of the tower cranes to obtain real-time on-site objects [34–36], especially static and dynamic objects on the ground.

2.2.2. Tower crane hoisting monitoring

Research on hoisting monitoring is mainly concerned with the health status of the tower crane to avoid some structural damage and the risk of buckling instability. For instance, Nadar designed a monitoring system of the tower crane based on wireless sensor network, which can collect the torque data of tower crane jib and conduct early warning of the overrun risk [37]. Li proposed a data-driven remote monitoring and early-warning system for tower cranes, which embeds a 3D kinematics model to provide threshold warning for the critical moment state of the structure [38]. Lalik realized the rapid measurement of the mast pressure state by the partial tilt angle of the tower crane body and jib [39]. Sayed analyzed the structure health state of the tower crane by establishing the relationship between its dynamic displacement and vibration characteristics [40].

2.2.3. Blind driving assistance and training

A large area of blind observation exists in the driver's line of sight for the larger-scale working space of the tower crane, while signalmen on-site are also unable to provide accurate information, thereby causing a great safety risk. Thus, researchers have built a global visual driving assistance system by various sensors and BIM. Drivers could obtain the hook motion and load [41], on-site invisible areas [42,43], and safety

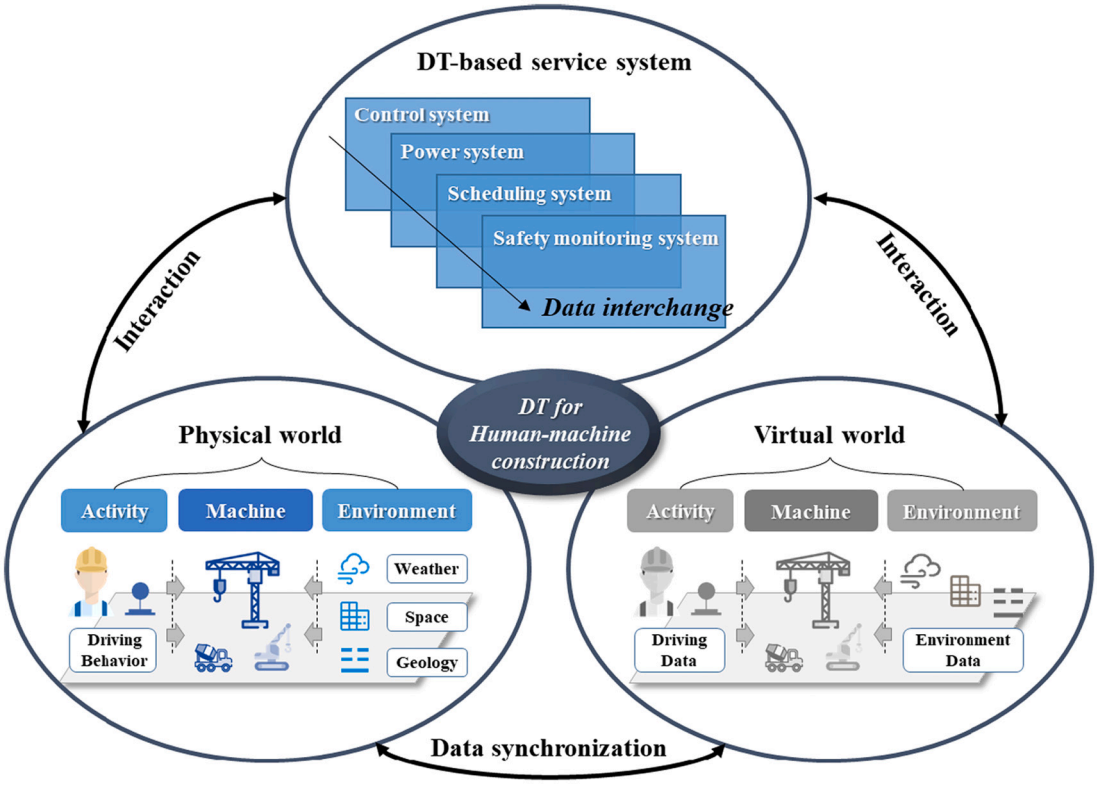


Fig. 1. Framework.

risk situations [44]. At the same time, some studies have presented the planning method of hoisting paths with two goals of safety and efficiency to actively guide the driver to control [45]. Other researchers have focused on drivers' fatigue, emotional states, and driving skills by using eye movement monitoring, facial monitoring, EEG monitoring, and VR to help drivers [46–48].

Tower crane safety is therefore an extremely important topic. From the point of view of Heinrich's rule [49], the existing studies have paid little attention to drivers' operational behaviors. According to the tower crane accident analysis and report [2–4], overload, tilt hoisting, and other unsafe behaviors are the fundamental reason of the structure damage, instability, and collapse of tower cranes. At present, effective monitoring means are lacking, and the influence of different driving behaviors on the stability of tower crane structure needs to be further explored.

2.3. Tower crane modeling and simulation

Modeling is an effective means to analyze the internal mechanism and operation process of construction machinery. The finite element method (FEM) is still considered to be the most effective method to analyze the structural mechanics and dynamics of the tower crane. Rauscher conducted the elastoplastic analysis of tower crane structures by using FEM in consideration of the dynamic couplings among steel structure, load motion, system drives, and rope [50]. Wang described the damage state and normal state model of the tower crane, which was distinguished by regression calculation of displacement data [51]. The scale model is another research method for structural analysis. Compared with the numerical model, the physical scale model can test the operation and control process with experiment, and it can easily find the problems and deviations in different working conditions. Kenan put forward the construction method of the tower crane scale model based on similarity theory and compared the structural modal analysis of the model under the scale factors of 30 and 40 [52]. Čápková studied the vibration damping of the tower in variable amplitude motion through a scale model [53]. Meanwhile, a large number of studies have focused on control optimization by using the scale model to study the double pendulum rule and the adaptive swing strategy for tower cranes [6,54]. All the abovementioned studies provide a foundation for the DT of the tower crane.

Tower crane simulation is another major research area. Ushio analyzed the failure modes of different mast joint types under the seismic load based on FEM [55]. Chen studied the dynamic effects on different parts of the tower crane (e.g., cat head, jib, counter-jib, and mast) under different total wind loadings from different angles, and the evaluation scheme of the total safety of the tower crane was proposed [56]. Voisin modeled the tower crane and its surrounding buildings and analyzed the transient effect of the tower crane foundation and auxiliary arm under strong wind condition [57]. Zrinic concluded based on FEM mechanical experiments that the failure of the angle brace plate of the counter-jib and the low quality of the weld are the main reasons for the structural failure of the tower crane [58]. The simulation results of Cho showed that the maximum deformation of tower crane hoisting is likely to occur at the jib-end, which needs to be enhanced during design [59]. Obviously, simulation helps engineers understand the performance of different parts of the tower crane under different load-up conditions. However, the current research is mainly used to support the design optimization, and few methods for safety risk performance analysis with different hoisting behaviors are available. Milazzo analyzed the complexity and risk of human–machine interaction in the construction process and believed that intelligent means would be an effective way to ensure man–machine interaction through investigation [60].

Overall, the research of tower crane modeling and simulation focuses on the structural steady state under environmental load, including wind, earthquake, and snow. Nearly no discussion on the stability performance of tower cranes under unsafe driving behaviors is available. However, construction practice has indicated that human driving behaviors are an important cause of safety risk accidents, and related research is

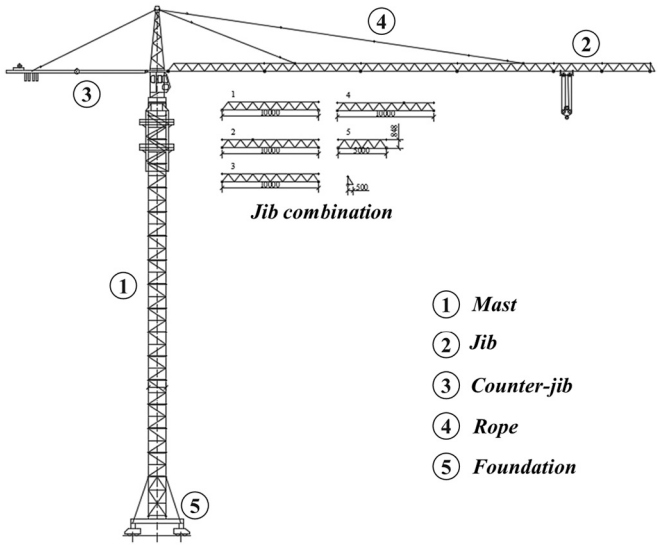


Fig. 2. Main structure of tower crane (e.g., XCMG QTZ80A-6010).

Table 1
Geometric similarity factors.

Structure	Scale model	Scale size (L, W, H; mm)	Real size (L, W, H; mm)	S_l
Mast		(160, 160, 160)	(1600, 1600, 1600)	1/10
Jib		L: 1200	L: 15000 (Two jibs)	1/ 12.5
Counter-jib		L: 600	L: 8000	1/ 13.3
Rope		L: 1430	L: 16500	1/ 11.5
Cross foundation		L*H: 600*600	L*H: 6000*6000	1/10

Table 2
Material similarity factors.

Attributes	Scale material (Aluminum alloy 6061)	Real material (Steel Q235)	S_ρ	S_e	S_m
Density	2.8E3(kg/m^3)	7.85E3(kg/ m^3)	0.36	/	/
Unit mass	2.8E3(kg)	7.85E3(kg)	/	0.36	/
Elastic Modulus	6.8E11(Pa)	20.6E11(Pa)	/	/	0.32

necessary. However, simulating and analyzing the unsafe driving behavior incentive on the actual large tower crane are dangerous. Thus, a research idea based on DT is proposed in this study to reflect the influence of unsafe hoisting behaviors on the stability of the total tower crane and its local components.

3. DT of tower crane

Any DT requires two components: the physical and virtual worlds. Data flow between them, transform, and generate service value. In the construction phase, DT for human–machine interaction activities consists of the driver, machinery, and environment. In this system, the state of machinery is the core because it directly determines the construction efficiency, safety, and progress. From the perspective of safety, this study describes the DT of human-machine interaction construction activities, as shown in Fig. 1. Among them, the construction machinery will be built as a virtual model, and the driving and environment data collected by sensors will be analyzed on the virtual model as the external excitation effect. This data interactive process will provide more accurate analysis services of operational efficiency, energy consumption, and operational safety. To comprehensively illustrate the framework, a tower crane is taken as the research object to propose the method and experiment of the DT for human–machine interaction activities, including three parts: tower crane scale model, virtual model, and data collection and flow based on IoT.

3.1. Scale components of tower crane

The scale model is the physical entity in the DT, which is an important way to replace the real large-scale structures for mechanical analysis. The similarity principle is the fundamental basis to establish a scale model, including geometric properties, material properties (density, elastic modulus, and unit mass), and external boundary conditions [61], as in Formula (1). When the geometric similarity factors (scale model: real model) are maintained at 1/30 or above, the scale model can show a more realistic dynamic response effect, such as structural modal analysis [52]. In the research, the key structural components of the tower crane are used as the main scale objects to form a scaled model, as shown in Fig. 2. The model includes mast, jib and counter-jib, rope, and foundation. Among them, the geometric similarity factor is approximately 1/10, as shown in Table 1. In terms of materials, the aluminum alloy is selected to ensure the experimental strength of special working conditions, and the material scale factor is approximately 1/3, as shown in Table 2.

$$\begin{cases} S_l = \dfrac{l_t}{l_p} = \dfrac{b_t}{b_p} = \dfrac{h_t}{h_p} \\ S_m = \dfrac{m_t}{m_p}, S_\rho = \dfrac{\rho_t}{\rho_p} \\ S_E = \dfrac{e_t}{e_p} \end{cases} \tag{1}$$

S_l- Geometric similarity factor, S_m- Mass similarity factor, S_ρ- Density similarity factor, S_e- Elastic modulus similarity factor, t- Scale model, p- Real model.

3.2. Tower crane virtual model

The virtual model is a virtual testbed for structural dynamic analysis, and operation safety analysis of the tower crane, as shown in Fig. 3. From the perspective of the hoisting process, the virtual model is a comprehensive synchronization of the actual hoisting conditions of the tower crane. Among them, hoisting behaviors (load size and direction), mechanical attitude (initial hoisting height, rotation angle, and radius change distance), and environment conditions (wind pulsation) act as real-time external incentive time series data on the virtual model. The virtual model will analyze the structure dynamic response, stability, and damage of the tower crane to realize the experiment simulation under different hoisting conditions.

The establishment of the virtual model is based on the size and hoisting condition of the scaled model, including the component, FEM, and BIM models. The component model is the structure and mechanism unit that makes up the scale tower crane, and Solidworks is used to model the complex mechanical components. The FEM model is the core of dynamic analysis in DT. At this point, the structural components are transferred as the different simple element units according to their stress statefor finite element calculation and analysis. Among them, mast, jib,

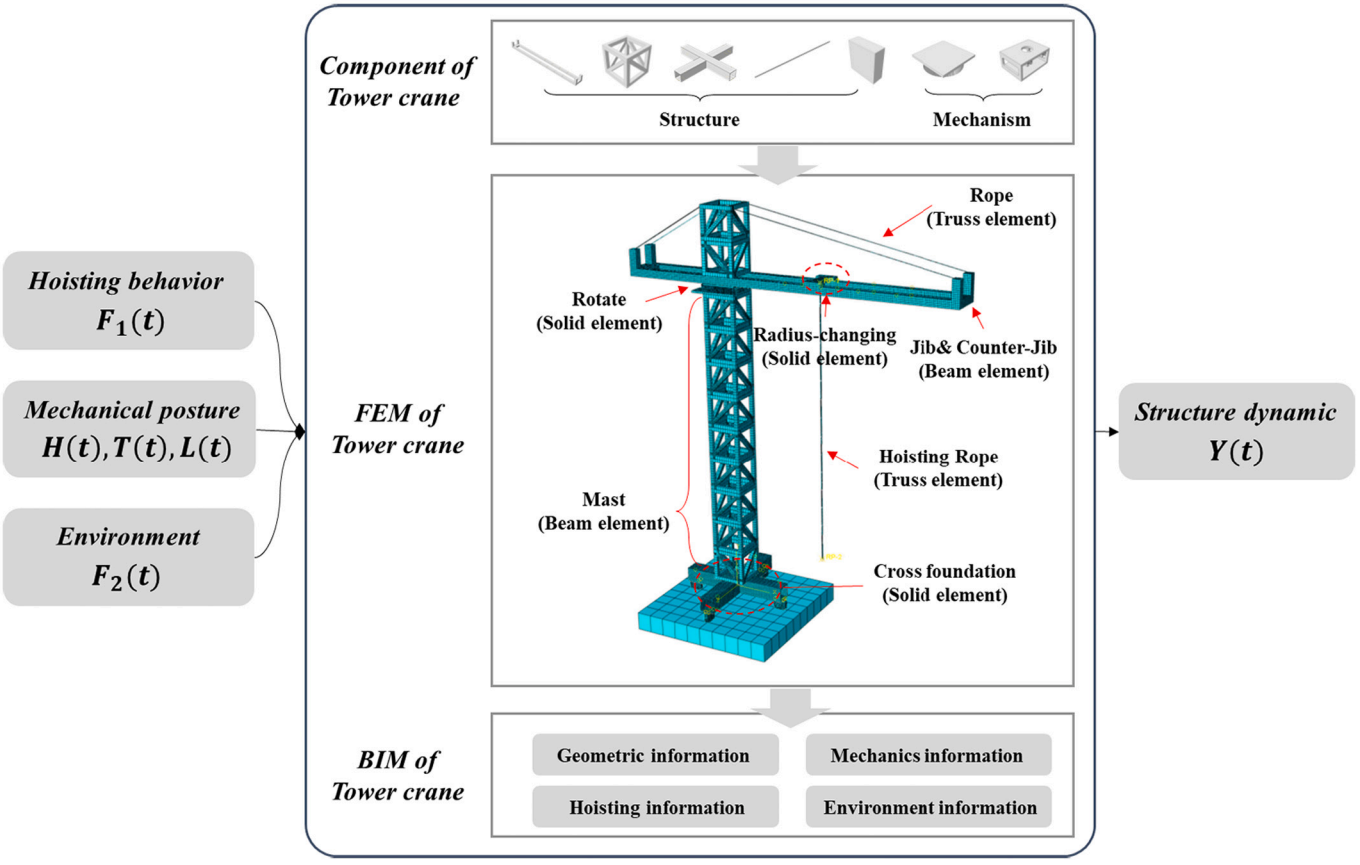


Fig. 3. Tower crane virtual model.

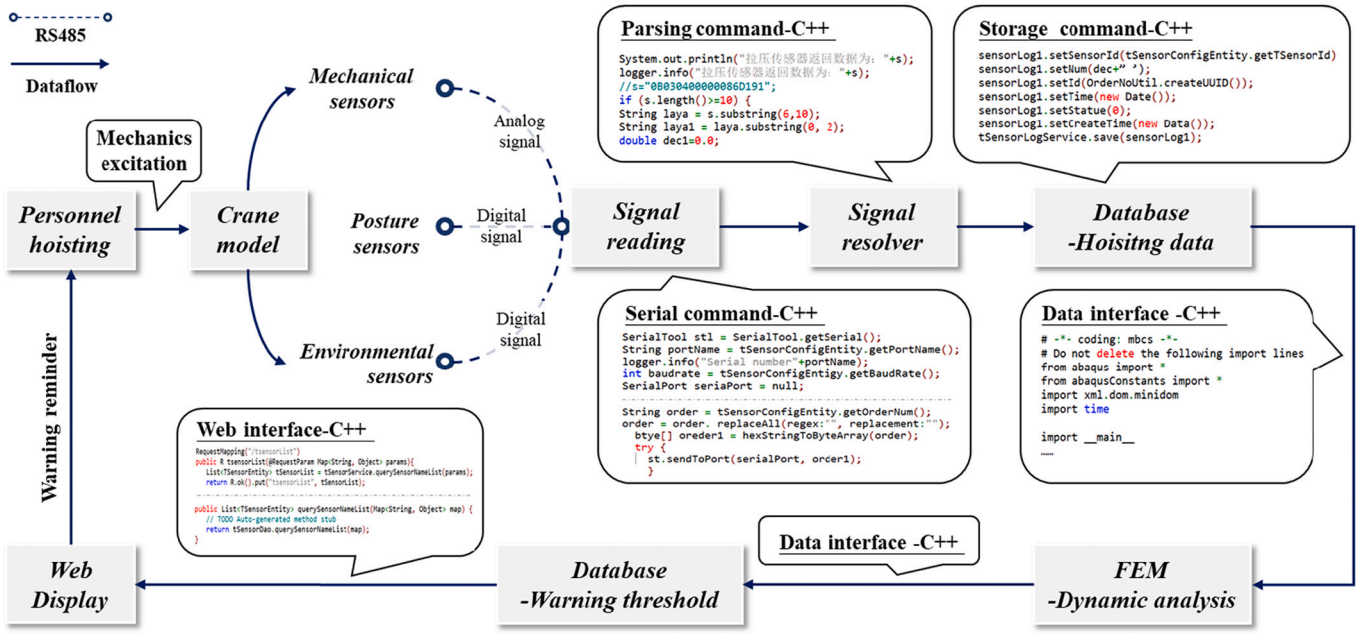


Fig. 4. Dataflow command.

and counter-jib belong to long and thin jib (length: sectional area > 10), which can be simplified by the beam unit. The rope can withstand tension but not stress and is simplified by the truss unit, and prestress is applied to the truss unit to ensure that the ropes are tensioned before hoisting. Rotating, radius-changing, and cross foundations are parts where the mass is concentrated, and they are simplified by the solid unit. The abovementioned method not only can effectively simplify the structural components on the basis of satisfying the refinement of structural analysis but also can speed up analysis when loading the different external excitations. BIM is used to store all the information of

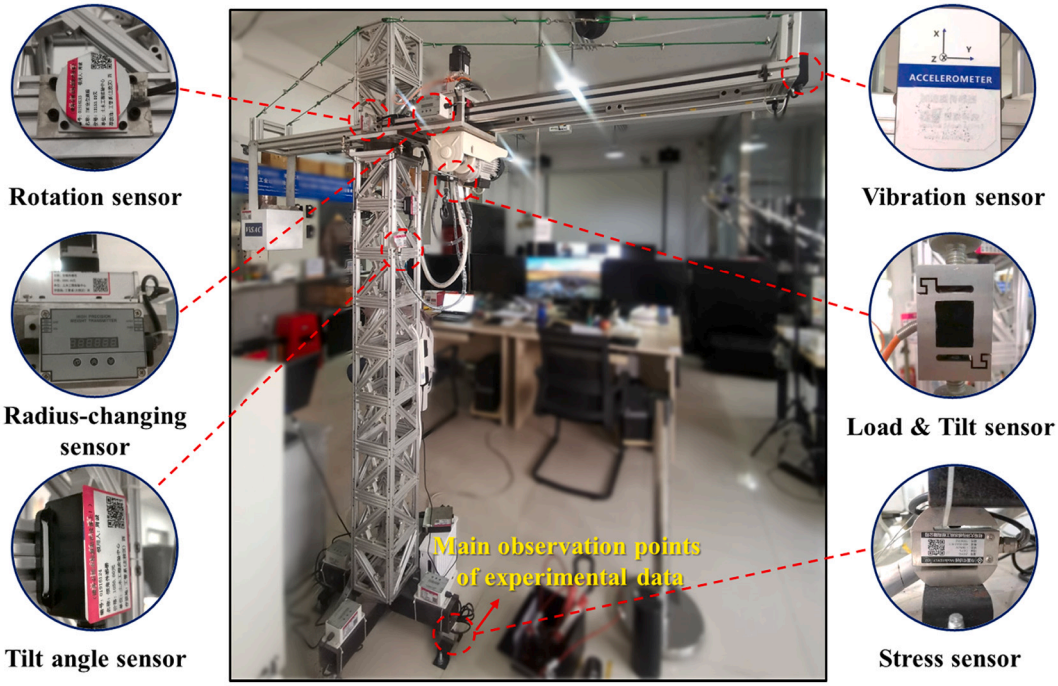


Fig. 5. Scale model with sensors.

Table 3
Basic parameters of sensors (20 Hz).

Sensor	Size (mm)	Range	Precision	Number	Position
Rotation	66 × 56 × 29	±360°	0.02°	1	Rotation
Radius-changing	45 × 89 × 149	0.1–80 m	±2 mm	1	Radius
Tilt-angle	66 × 56 × 29	±80°	0.02°	2	Mast body and hoisting rope
Vibration	66 × 56 × 29	±200 g	±0.1 m/s^2	1	Jib-end
Load	45 × 89 × 149	0–200 kg	0.01 kg	1	Hoisting rope
Stress	45 × 89 × 149	0–300 kg	0.01 kg	4	Foundation

the DT-driven hoisting experiment, including the geometry of the scale components, mechanics information, hoisting, and environment information.

3.3. DT dataflow in hoisting operations

IoT is the foundation of the virtual and real synchronization in the DT. According to the hoisting characteristics of the tower crane, collecting mechanical, posture, and environmental data during the operation is necessary. Among them, the mechanical data reflect the load state, local stress, and vibration of the structure. The posture data reflect the operation state of the tower crane and indirectly reflect the structure form of the tower crane when hoisting, including the position of circumrotating and radius-changing. Environmental data specifically refer to wind load, which is the most easily disturbed load for tower cranes and other tall structures.

In the process of human–machine interaction, the data go through a completed closed-loop flow from personnel hoisting, sensor collection, dynamic analysis, and danger warning, as shown in Fig. 4. First, the data from different sensors are numbered according to the serial port and asked to read data. Then, the parsing command will read analog and digital signals, and the key information of different sensors, such as ID, data acquisition time, and value, are all stored in the database. Next, the interface program of the finite element input file of ABAQUS converts the sensor data in the database into effective excitation and loads it on the virtual model to realize the dynamic analysis. Finally, the results are synchronized to the database, and a structural dynamic threshold under conventional operations is set as the warning benchmark. Notably, once the real-time analysis result exceeds the threshold, the hoisting behavior will be identified as dangerous, and the warning information will be displayed on the web interface to remind personnel to correct it.

4. Tower crane hoisting experiment

4.1. DT experiment platform

The tower crane scale model is made and assembled before the hoisting experiment based on the similarity theory. Meanwhile, the sensor and monitoring system is installed on the scale model simultaneously, which covers each component of the tower crane, as shown in Fig. 5. This experiment mainly focuses on the dynamic effect analysis of the tower crane under different hoisting behaviors and temporarily ignores the influence of wind load. Therefore, the sensors are mainly composed of rotation, radius-changing, jib vibration, hoisting rope load, mast body tilt-angle, and foundation stress. Foundation stress data are used as the main observation data to compare with the simulation data in the experiment. Table 3 shows the sensor parameters in detail. In the terms of software, C++ is used for system development deployed in Windows 10 to support real-time data collection and online display on the web. Each sensor is programmed according to the industrial 485 communication protocol. The database is developed by the Navicat platform and MYSQL.

The hoisting operation is monitored in real time based on the DT experimental platform. The human–machine interaction interface is shown in Fig. 6, including control platform, monitoring platform, and tower crane. Among them, the control platform has two modes of manual and automatic. The three-degree-of-freedom control of lifting, turning, and luffing can be realized through the touch screen. At the same time, it has the safety protection of emergency braking in

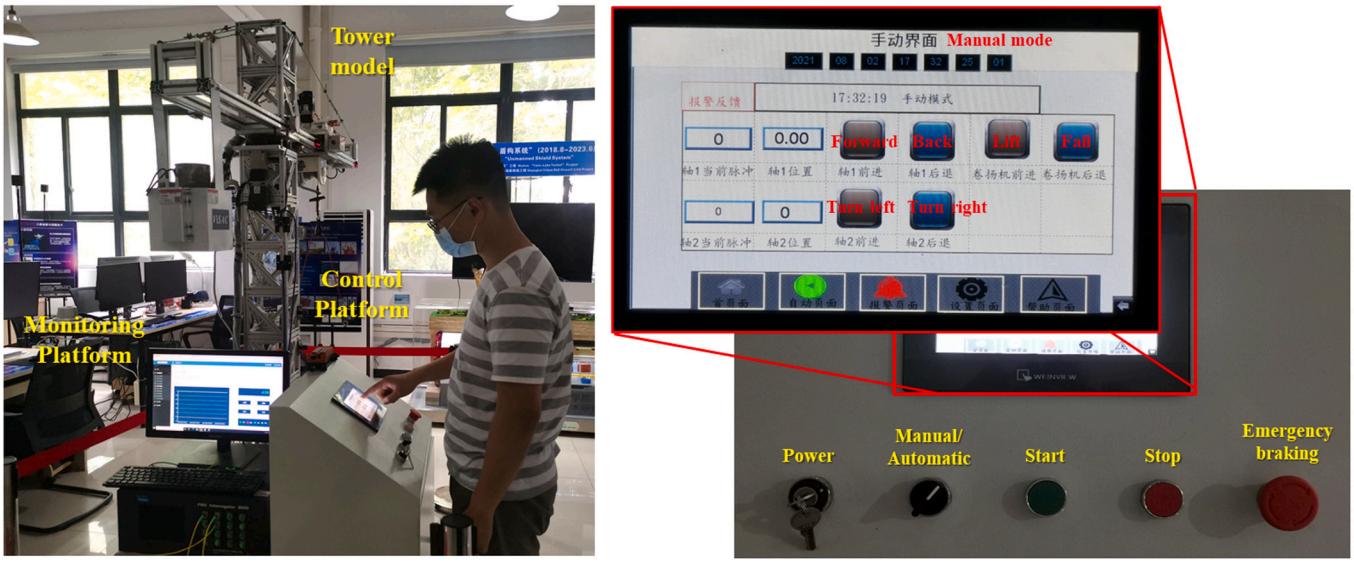


Fig. 6. Control interface in DT.

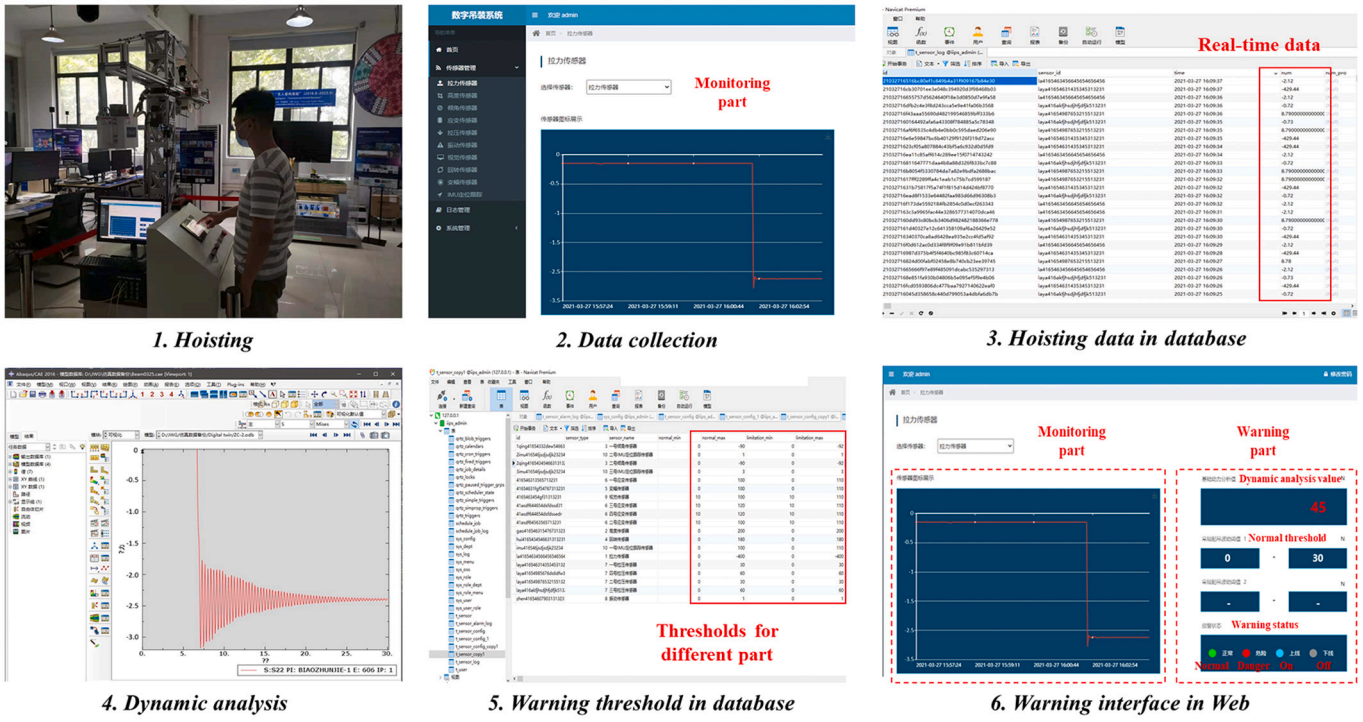


Fig. 7. Dataflow in the experiment.

Table 4
Experimental planning and basic parameter.

Tower crane posture	Rotation	Luffing	Hoisting height	Load
Parameters	0°	250 mm	1700 mm	2 kg

Hoisting condition	Normal	Hoisting rope tilt	Braking time	Unload time
Parameters	/	10°	0.2 s	0.2 s

dangerous situations. As shown in Fig. 7, real-time interactive data will be collected by the monitoring system and stored in the database for each hoisting operation activity, and then, they will be used for dynamic analysis. Subsequently, the results will be exported to the database, and the unsafe hoisting behavior will be warned by the warning threshold, and the corresponding alarm information will be synchronized to the interface of the monitoring system to prompt the operator.

4.2. Hoisting experiment condition design

Experiment conditions compose the most common operation behaviors in the hoisting process: vertical hoisting, tilt hoisting, sudden braking, and sudden unloading. To ensure the validity of experiments,

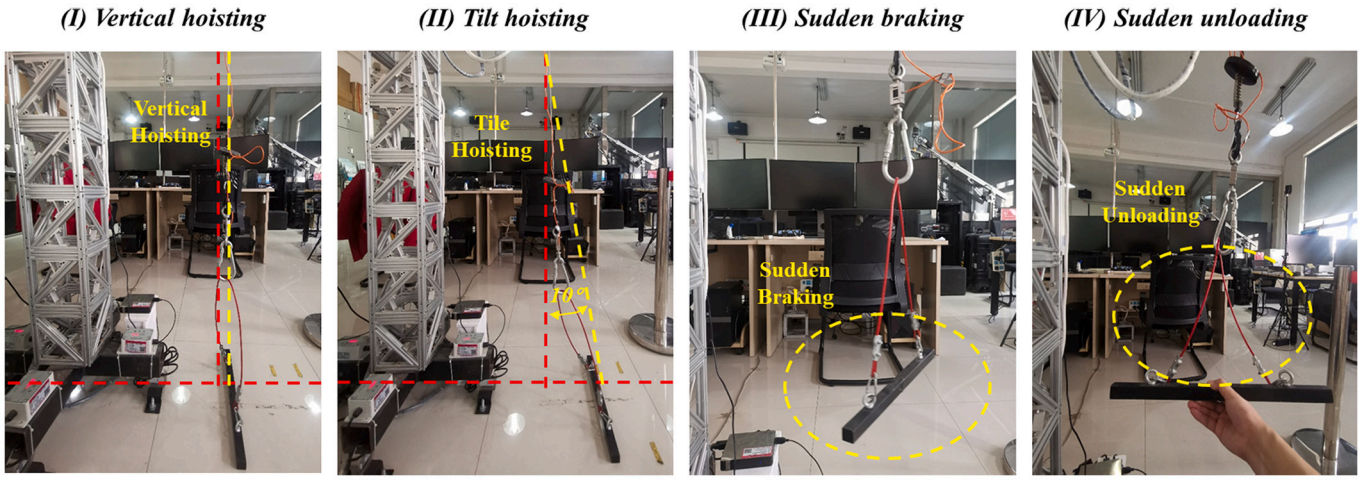


Fig. 8. Four hoisting behaviors in experiment condition.

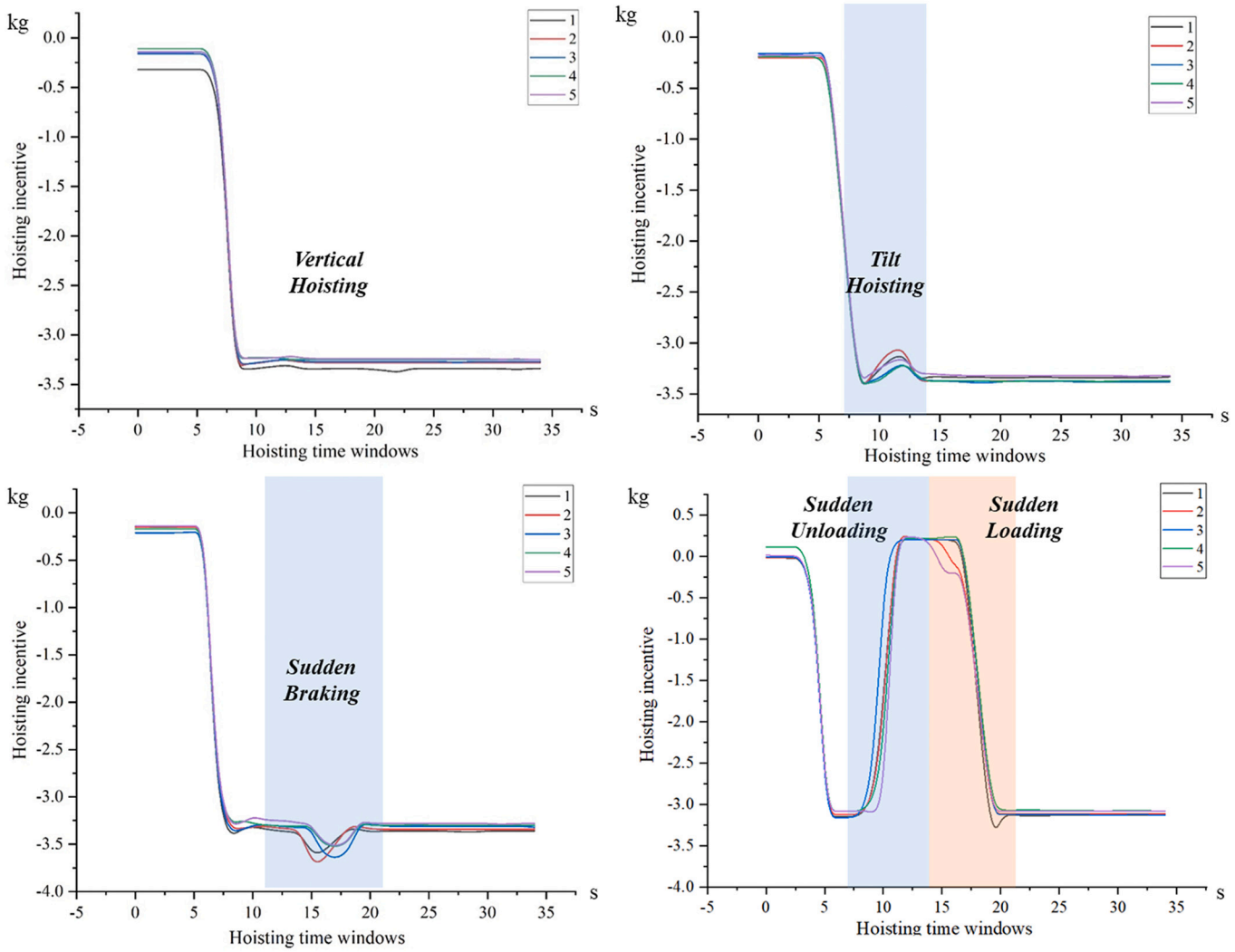


Fig. 9. Load excitation of four hoisting behaviors.

the abovementioned hoisting behaviors are more clearly defined according to the general management regulation in the construction site. (i) Vertical hoisting is when the suspender, hook, and lifting mechanism keep in a straight line and are perpendicular to the ground. (ii) Tilt hoisting is when the line where the suspender, hook, and lifting mechanism are located is not perpendicular to the ground; from the perspective of physics, tilt hoisting is when the angle greater is than 5°. (iii) Sudden braking means that the hoisting speed is instantly reduced

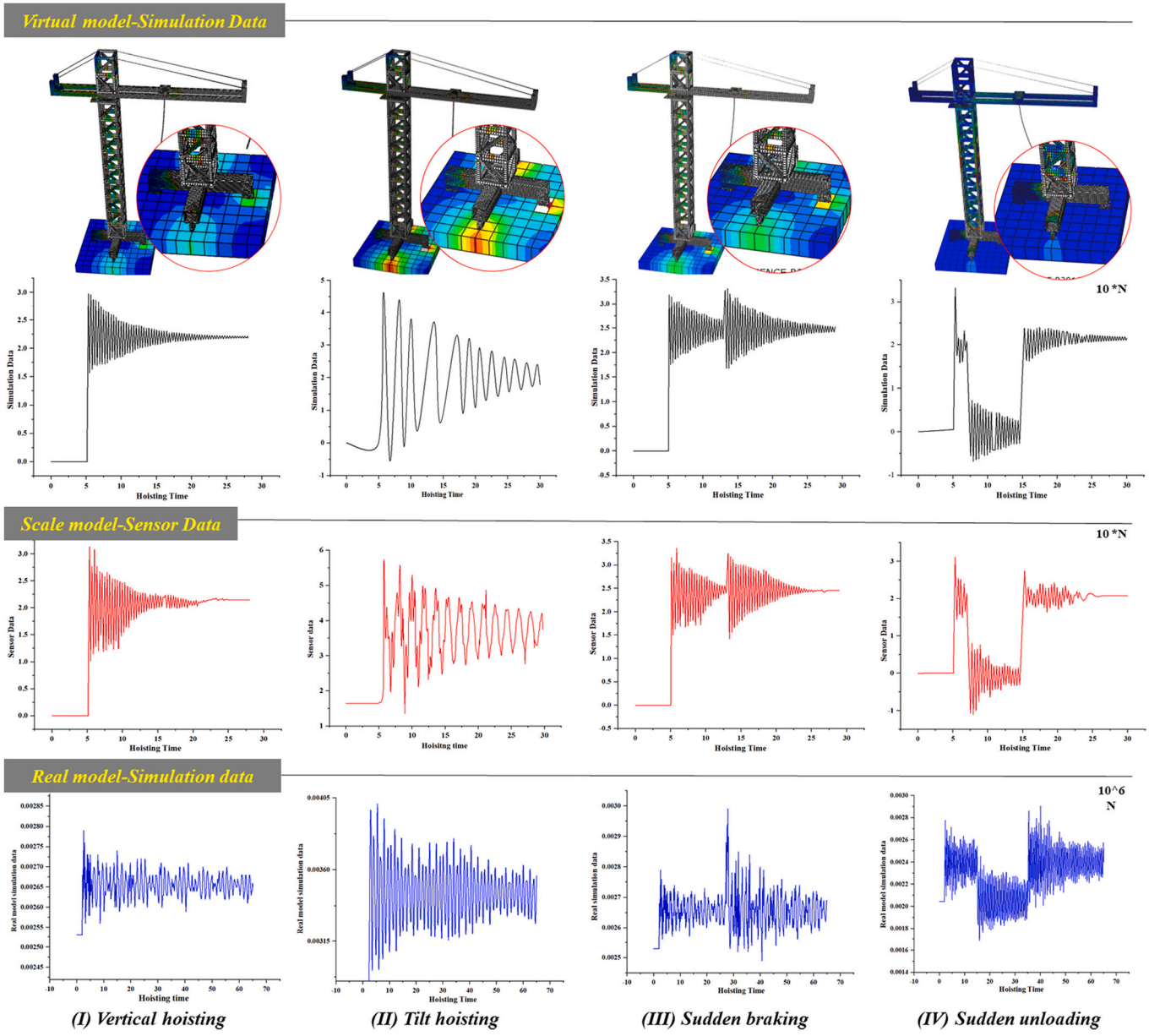


Fig. 10. Dynamic response results of four hoisting behaviors.

Table 5
Total collapse level in incremental load steps.

Incremental Steps	Vertical Hoisting	Tilt Hoisting	Sudden Braking	Sudden Unloading
1	0.05205	0.05767	0.05674	0.05379
15	0.39503	0.4707	0.46062	0.4075
30	0.76283	0.91384	0.89079	0.78632
45	1.13247	1.35659	1.32198	1.16533
60	1.54042	1.61466	1.66712	1.54319
75	1.70668	1.71771	1.77574	1.72025
90	1.80428	1.80133	1.87792	1.8198
105	1.89413	1.8816	1.98032	1.91253
120	1.98178	1.97888	2.08178	1.9905
135	2.06541	2.01255	2.13335	2.07745
150	2.16181	2.03021	2.17402	2.10412

to 0 during hoisting. (iv) Sudden unloading is when the suspender is lost due to unfirm binding or other reasons, and the load becomes 0. The basic parameters of the experiment are designed according to the abovementioned four hoisting behaviors, including the tower crane posture parameters and the hoisting behavior parameters, as shown in Table 4. Meanwhile, each experiment will be repeated five times for analysis and the time window is kept to 35 s to ensure the result accuracy. During the experiment, the suspender (beam) will be placed on the initial position marked in advance, and the tower crane will be controlled by the researchers to complete the experiment of different hoisting behaviors, as shown in Fig. 8.

4.3. Hoisting experiment result

4.3.1. Load excitation of the hoisting behaviors

In related works, more studies consider the load excitation of wind, snow, and earthquake for tower cranes. In this experiment, the load excitation effects of different hoisting behaviors on tower cranes are summarized through DT, as shown in Fig. 9. The load excitation peak and amplitude of tilt hoisting are larger than those of vertical hoisting. Sudden braking will produce strong transient effects during hoisting. The amplitude of load variation is obvious in the process of sudden

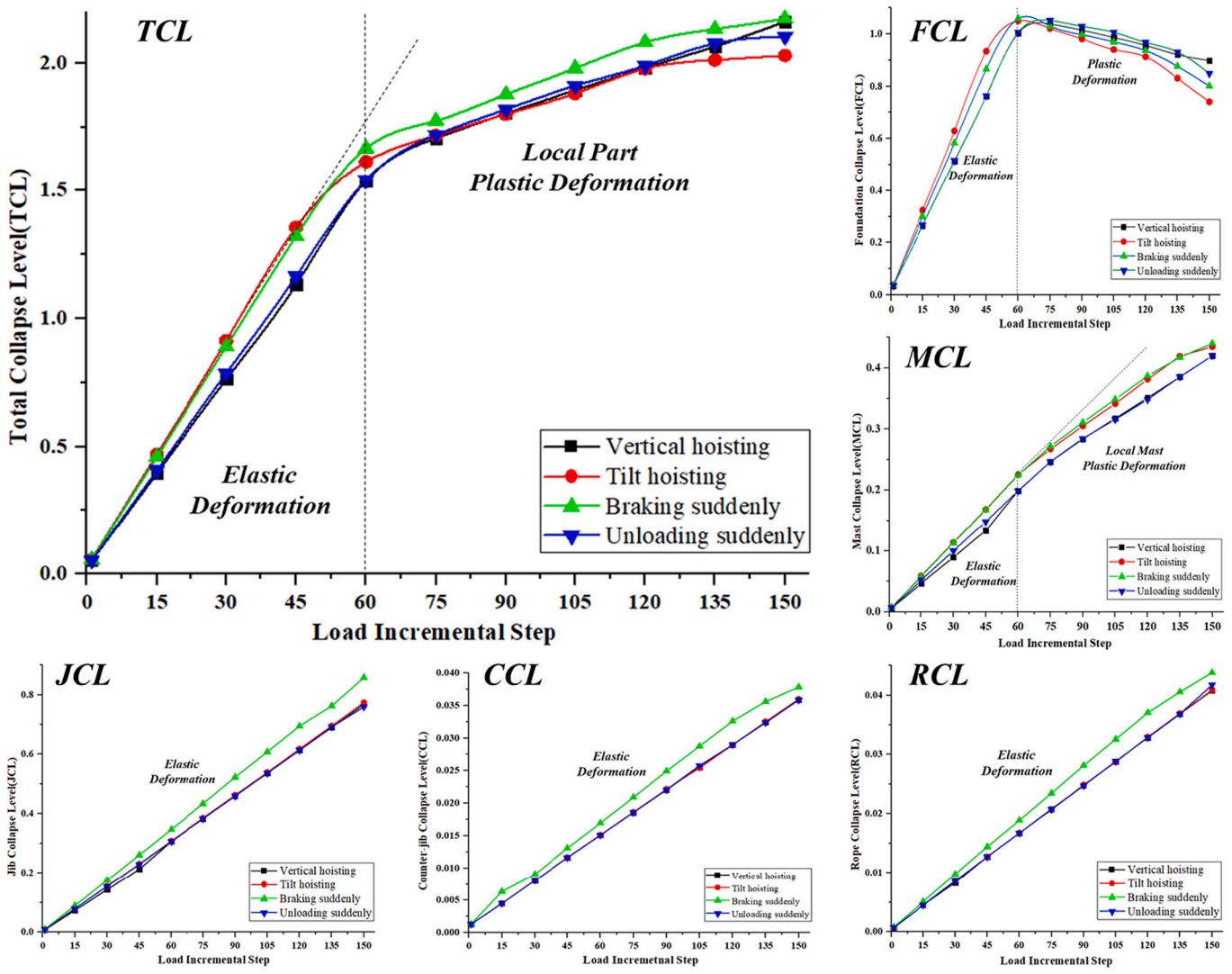


Fig. 11. Stability analysis in load incremental steps.

unloading and loading. The abovementioned load excitation data will be the input data for the virtual model in DT to obtain the structural dynamic results of the tower crane.

4.3.2. Dynamic response results of the tower crane

Taking the data of observation points as an example (Fig. 5), the dynamic response results of the tower crane under four hoisting behaviors are obtained simultaneously in the virtual model. First, we compare the sensor data of the scale model, the simulation data of the virtual model, and the simulation data of the real model to ensure the accuracy of the results, and the analysis frequency is 2 Hz. The load is scaled proportionally, and the specific parameters refer to Tables 1 and 2. As shown in Fig. 10, the sensor data of the scale model are consistent with the simulation data of the virtual model, which proves the usability of the virtual model in DT. Meanwhile, compared with the simulation data of the real model, the data of the scaled model are the same in the time series trend but differ in the data magnitude. This phenomenon is due to the difference in the material properties of the structure, but it does not affect the analysis of the dynamic effects under different hoisting behaviors in this experiment.

The trend of the data shows that the significant difference in time series data is caused by the four different hoisting behaviors. The dynamic response caused by the tilt hoisting is larger and the reduction cycle is longer than those by the vertical hoisting. Given the obvious change in acceleration, the sudden braking produces the obvious transient effect, which enlarges the original dynamic response amplitude. During the process of sudden unloading to loading, the dynamic response data show a significant downward and upward trend with the changes in hoisting load. Overall, the time series data of dynamic response under different unsafe behaviors are obviously different, and the influence on the structural stability will also be different. At the same time, different data trends also provide the basis for classifying and identifying unsafe hoisting behaviors.

In addition, the equivalent stress cloud of the virtual model can intuitively reflect the dynamic response effect of different hoisting behaviors when they happened. When tilt hoisting and braking occur, the stress of the cross foundation increases significantly, and the effect of the tilt hoisting is stronger. When sudden unloading occurs, the stress of the foundation decreases significantly.

5. Stability analysis and discussion

5.1. Stability analysis method based on incremental dynamic load

Different hoisting behaviors will inevitably affect the stability of the tower crane structure. From the perspective of material mechanics, the instability of the tower crane is due to that the structural stress is greater than its yield stress, which results in plastic deformation. Thus, based on

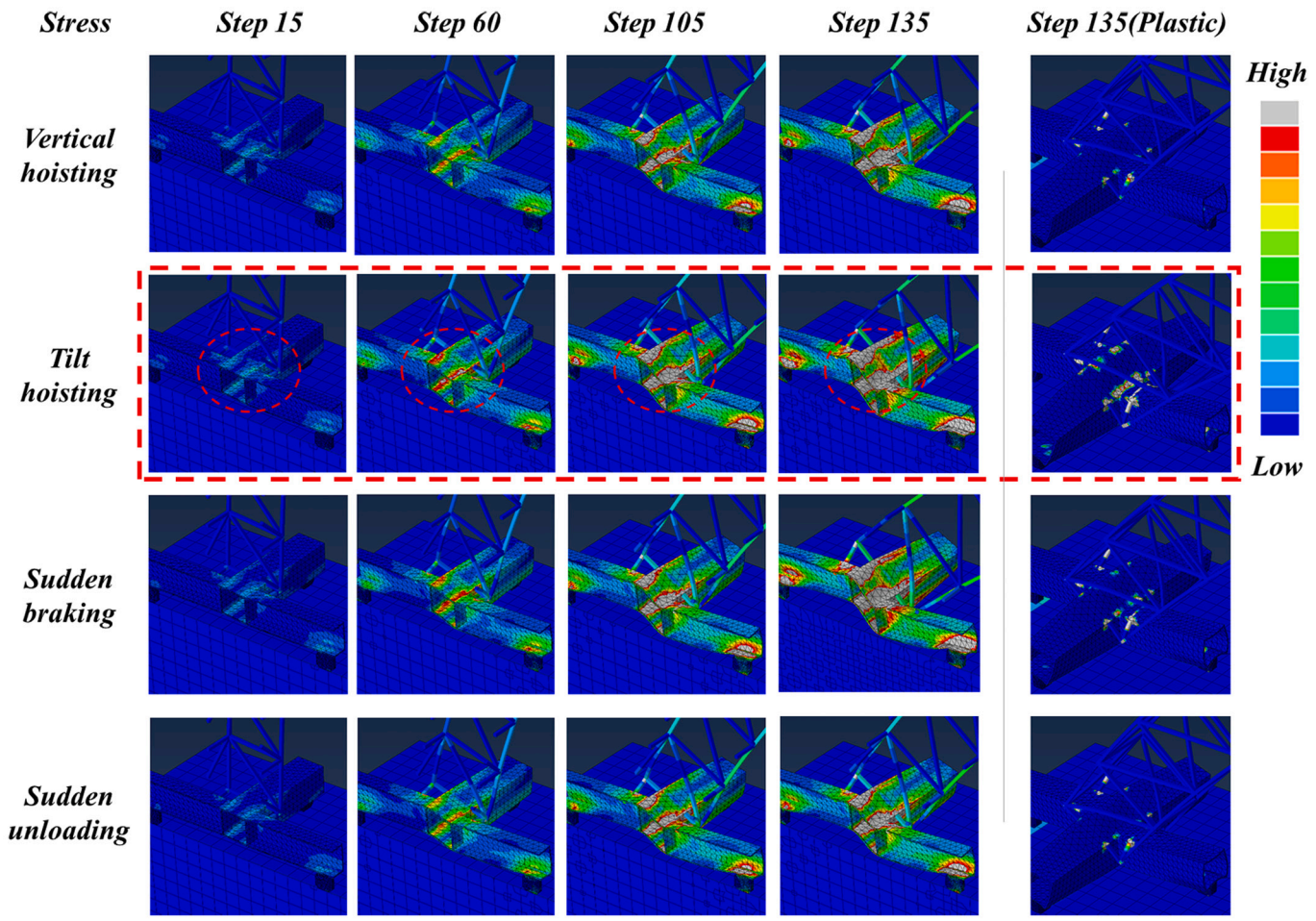


Fig. 12. Foundation instability evolution under four hoisting behaviors.

the maximum equivalent stress criterion in the general steel structure failure analysis method [62], this study proposes a collapse probability level C_i to reflect the stability of the structure component unit, as shown in Formula (2).

$$C_i = \frac{M_i}{M_q} \tag{2}$$

M_i- Element maximum equivalent stress data;
M_q- Element yield stress.

A method to describe the collapse level of different components is established based on the unit composition, as shown in Formula (3), to simultaneously consider the stability differences of the foundation, mast, jib, counter-jib, and rope. When the collapse level for each component is greater than 1, the structure will undergo plastic deformation, which results in the occurrence of instability accidents. The total collapse level is described by Formula (4).

$$\{FCL, MCL, JCL, CCL, RCL\} = \sum_1^n \frac{C_i}{n} \tag{3}$$

$$TCL = FCL + MCL + JCL + CCL + RCL \tag{4}$$

TCL-Total collapse level; FCL-Foundation collapse level;
MCL-Mast collapse level; JCL-Jib collapse level;
CCL-Counter-jib collapse level; RCL-Rope collapse level.

Based on all the above mentioned collapse levels, the research refers to the incremental dynamic analysis (IDA) method [63], which can effectively describe the process of the total tower crane and local component units being loaded to the instability under different hoisting behaviors. The incremental load analysis relies on the DT, the experimental load excitation is magnified by 15 times, and 10 loading steps are established. The results of plastic time history analysis reflect the different collapse levels of each component in the process from elastic deformation to plastic deformation for analyzing the stability state under different hoisting behaviors.

5.2. Stability discussion of the tower crane

5.2.1. Stability changes in load incremental steps

The first thing we can discuss is the stability changes caused by different hosting behaviors during the load incremental step process, as shown in Table 5 and Fig. 11. For the TCL, the collapse risk of the tower crane gradually slows down after a rapid increase as the load increases. This result is caused by plastic deformation in local parts. In the process of elastic-plastic deformation, the effect of tilt hoisting and sudden braking on the stability is increasingly becoming obvious, and the growth and declining rate are also faster. Meanwhile, tilt hoisting is the first hoisting condition that produces plastic deformation, which means that it is most likely to cause local component instability. The rigidity of FCL has been significantly degraded during the incremental process. Among them, tilt hoisting has also the greatest effect on the stability of the foundation. In addition, the increased amplitude in MCL has also changed, which indicates that slight plastic deformation has also occurred in the mast part of the tower body. Although JCL has not undergone a change in growth rate, it maintains at a relatively high level. By contrast, CCL and RCL are maintained in a relatively stable state, and the effect of instability is low. In summary, all unsafe hoisting behaviors, especially tilt hoisting and sudden braking will bring obvious influence

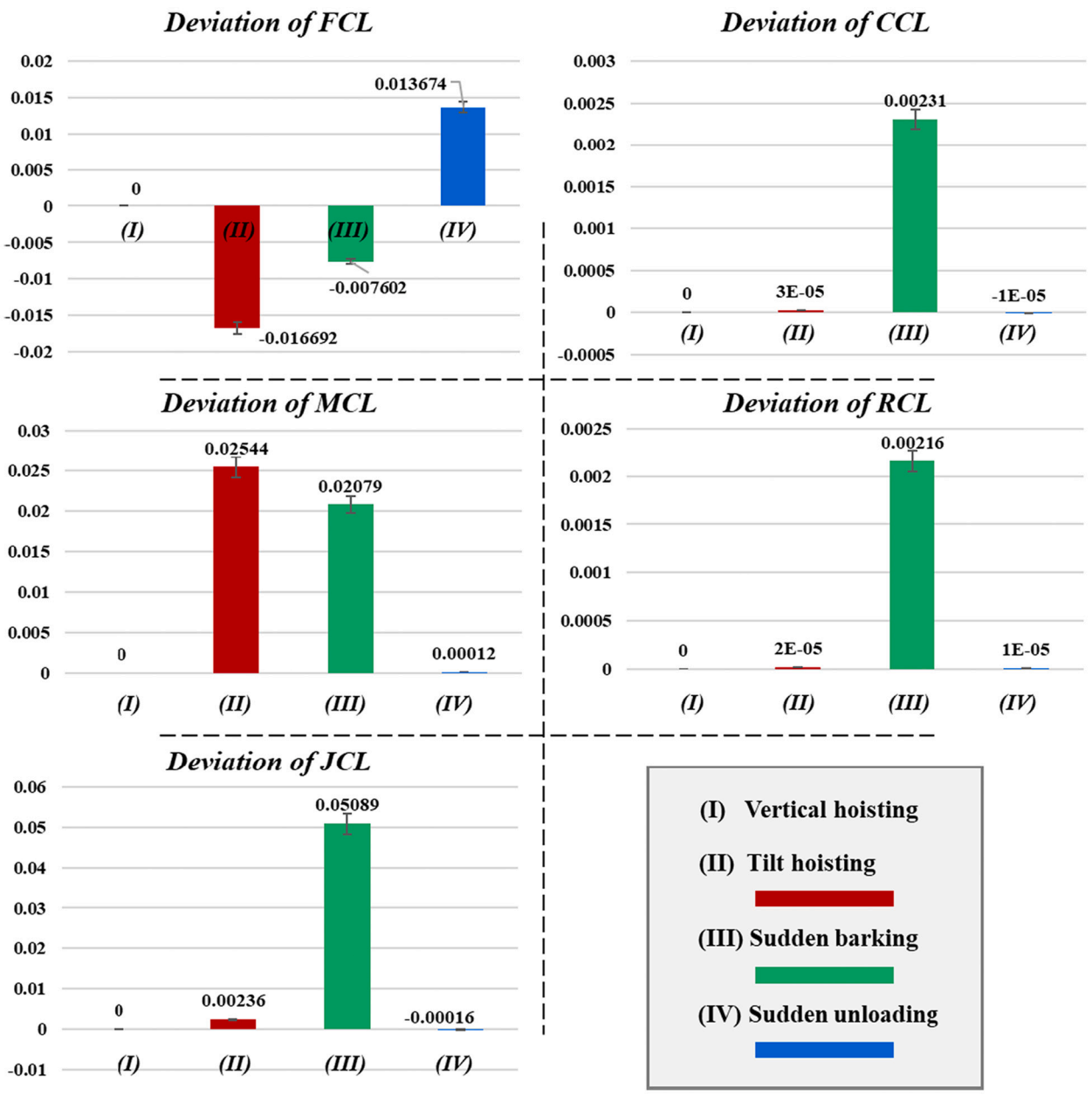


Fig. 13. Stability characterization of structural components (the 60th step).

to the stability of the tower crane. However, the existing literature mostly use the structural threshold as the criterion to determine the safety state, but the instability risk often occurs when the structure is damaged. Therefore, unsafe hoisting behaviors should be monitored, which are the root cause of accelerated structural damage. In this way, the monitoring system can predict the risk of instability more accurately and earlier.

5.2.2. Foundation instability evolution under hoisting conditions

Aiming at the cross foundation where the instability occurs first, we can further observe the separation process of the foundation unit under different hoisting conditions, as shown in Fig. 12 (stress-section view, plastic-total view, and deformation magnification for 50 times). The tilt hoisting and sudden braking have already undergone more obvious plastic deformation at the incremental **Step 135 (Plastic)**. From the perspective of the instability process, the damage for the tilt hoisting, which is the most severely damaged, first appears on the connection node of the foundation, especially the node where the side of the jib is located. Then, the damage gradually spreads to the two axes of the cross foundation. Finally (at step 135), obvious plastic deformation appears at the intersection of the foundation, and the deformation on the same side of the jib will also be more obvious. The abovementioned phenomenon can represent the instability propagation process of a typical hoisting behavior. If the same type of instability evolution process (e.g., crack) is found in the actual tower crane inspection process, then the engineer can concentrate on checking the driver's hoisting behavior during the hoisting process.

5.2.3. Stability characterization of structural components

With the 60th load incremental step in Fig. 11 as the object, the collapse index deviation of the unsafe hoisting conditions relative to the vertical hoisting is observed for different structural components, as shown in Fig. 13. For the cross foundation, the dynamic effect of the tilt hoisting is the greatest, and the effect of sudden unloading is significantly higher than the normal state. For the mast, the effect of tilt hoisting and sudden braking is similar, but the effect of sudden

unloading is not obvious. For the jib, the effect of sudden braking is the largest, and the effect of the tilt hoisting is similar to that of sudden unloading. For the counter-jib and the rope, sudden braking also has the greatest effect. Therefore, the distinction between different unsafe hoisting behaviors is better when the fluctuation of the data is more obvious. Thus, the dynamic response of the cross foundation or adjacent mast is the most sensitive and can be used as the main placement of the sensor when considering monitoring unsafe hoisting behaviors.

Tower crane safety is an important topic. Many interesting works can be explored in the future on the basis of this study. (i) The combined effect of environmental load and more unsafe hoisting behaviors on tower cranes can be discussed. (ii) The structural dynamic effect when the tower crane structure changes can be analyzed. (iii) The weight of the influence for different structural components during the instability process of the tower crane can be explored.

6. Conclusion

Human-machine interaction construction activities, especially for tower crane hoisting, have a very high safety risk, where unsafe hoisting behaviors can easily lead to accidents such as collapse and instability. Thus, this study proposes a general framework based on DT for the human–machine interaction construction activities. This framework is used to analyze the influence of the unsafe hoisting behaviors on the stability of the tower crane. In our research, a refined scale model is assembled based on the similarity principle, including cross foundation, mast, jib, counter-jib, and rope. The virtual model of the tower crane is constructed synchronously to cover the components, FEM, and building information. The component elements are divided by a reasonable strategy to ensure calculation efficiency and accuracy. Meanwhile, a data flow for real-time dynamic analysis of the tower crane is proposed based on IoT, which realizes the interaction between the virtual and physical worlds.

In the hoisting experiment, the time series effect of load excitation with vertical hoisting, tilt hoisting, sudden braking, and sudden unloading are summarized. The dynamic response characteristics are described under the four hoisting behaviors through the comparison of the tower crane scale model sensor data, simulation data, and the real tower crane model simulation data. Meanwhile, this study thoroughly discusses the effect of unsafe hoisting behaviors on the total and local stabilities of the tower crane, constructs the collapse probability function, and proposes the description methods by TCL, FCL, MCL, JCL, CCL, and RCL. The elastoplastic changes of each component under different working conditions is discussed based on the IDA method. The results show that tilt hoisting is the unsafe behavior that is most likely to cause tower crane instability, and the effect gradually becomes larger as the load increases. The foundation is the most sensitive part to unsafe hoisting behaviors, which can reflect instability characteristics in the tower crane monitoring. The research results provide a basic theoretical basis for the monitoring of unsafe hoisting behaviors of tower cranes in the future. The dynamic response effects of tower cranes under more complex environmental and behavioral coupling can be further explored in the future research.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgment

This research is supported in part by the National Natural Science Foundation of China (Grant Nos. 72171092, 71732001, U21A20151, and 71821001). This work is also supported by the Major Science & Technology Project of Hubei. China (Grant No. 2020ACA006). Sincere thanks to SRT partners, Yuchen He, Jinyan Han, Liangliang Qiang, and Weizhao Li, thank you all!

References

[1] J.D. Wiethorn, An Analytical Study of Critical Factors of Lift Planning to Improve Crane Safety Based on Forensic Case Studies of Crane Accidents (Doctoral dissertation), The University of Texas at Austin, 2018. repositories.lib.utexas.edu/handle/2152/67528.
[2] I. Shin, Jae, factors that affect safety of tower crane installation/dismantling in construction industry, Saf. Sci. 72 (2015) 379–390, https://doi.org/10.1016/j.ssci.2014.10.010.
[3] V. Tam, I. Fung, Tower crane safety in the construction industry: a Hong Kong study, Saf. Sci. 49 (2) (2011) 208–215, https://doi.org/10.1016/j.ssci.2010.08.001.
[4] P. Swuste, A 'normal accident' with a tower crane? An accident analysis conducted by the Dutch Safety Board, Saf. Sci. 57 (2013) 276–282, https://doi.org/10.1016/j.ssci.2013.03.002.
[5] M. Mains, J.R. Blough, Topics in Modal Analysis & Testing Vol. 10, Springer International Publishing, 2017, https://doi.org/10.1007/978-3-319-30249-2.
[6] H. Chen, Y. Fang, N. Sun, An adaptive tracking control method with swing suppression for 4-DOF tower crane systems, Mech. Syst. Signal Process. 123 (2019) 426–442, https://doi.org/10.1016/j.ymssp.2018.11.018.
[7] M. Schluse, L. Atorf, J. Rossmann, Experimentable digital twins for model-based systems engineering and simulation-based development, in: 2017 Annual IEEE International Systems Conference, 2017, pp. 1–8, https://doi.org/10.1109/SYSCON.2017.7934796.
[8] M. Schluse, M. Priggemeyer, L. Atorf, J. Rossmann, Experimentable digital twins streamlining simulation-based systems engineering for industry 4.0, IEEE Trans. Indust. Inform. 14 (4) (2018) 1722–1731, https://doi.org/10.1109/TII.2018.2804917.
[9] C. Boje, A. Guerriero, S. Kubicki, Y. Rezgui, Towards a semantic construction digital twin: directions for future research, Autom. Constr. 114 (2020), 103179, https://doi.org/10.1016/j.autcon.2020.103179.
[10] L. Zhang, J. Kang, H. Luo, B. Zhong, Drivers' physiological response and emotional evaluation in the noisy environment of the control cabin of a shield tunneling machine, Appl. Acoust. 138 (2018) 1–8, https://doi.org/10.1016/j.apacoust.2018.01.023.
[11] W. Jiang, L. Ding, C. Zhou, Cyber physical system for safety management in smart construction site, Eng. Constr. Archit. Manag. 28 (3) (2020) 788–808, https://doi.org/10.1108/ECAM-10-2019-0578.
[12] L. Hou, S. Wu, G.K. Zhang, Y. Tan, X. Wang, Literature review of digital twins applications in construction workforce safety, Appl. Sci. 11 (1) (2020) 339, https://doi.org/10.3390/app11010339.
[13] Y. Xu, Y. Sun, X. Liu, Y. Zheng, A digital-twin-assisted fault diagnosis using deep transfer learning, IEEE Access. 7 (2019) 19990–19999, https://doi.org/10.1109/ACCESS.2018.2890566.
[14] J.A. Erkoyuncu, I.F. del Amo, D. Ariansyah, D. Bulka, R. Roy, A design framework for adaptive digital twins, CIRP Ann. 69 (1) (2020) 145–148, https://doi.org/10.1016/j.cirp.2020.04.086.
[15] S.P. DiMaio, S.E. Salcudean, C. Reboulet, S. Tafazoli, K. Hashtrudi-Zaad, A virtual excavator for controller development and evaluation, in: Proceedings. 1998 IEEE International Conference on Robotics and Automation (Cat. No. 98CH36146) 1, 1998, pp. 52–58, https://doi.org/10.1109/ROBOT.1998.676255.
[16] N. Alaei, E. Kurvinen, A. Mikkola, A methodology for product development in mobile machinery: case example of an excavator, Machines. 7 (4) (2019) 70, https://doi.org/10.3390/machines7040070.
[17] K.Y.H. Lim, P. Zheng, C.H. Chen, L. Huang, A digital twin-enhanced system for engineering product family design and optimization, J. Manuf. Syst. 57 (2020) 82–93, https://doi.org/10.1016/j.jmsy.2020.08.011.
[18] D. Thomas, A. Hristo, M. Patrick, J.C. Göbel, F. Sven, A holistic system lifecycle engineering approach-closing the loop between system architecture and digital twins, Procedia CIRP. 84 (2019) 538–544, https://doi.org/10.1016/j.procir.2019.04.257.
[19] V.V. Zhidchenko, H. Handroos, A.N. Kovartsev, Fatigue life estimation of hydraulically actuated mobile working machines using internet of things and digital twin concepts, J. Phys. Conf. Ser. 1369 (4) (2019), 042025, https://doi.org/10.1088/1742-6596/1368/4/042025.
[20] V. Zhidchenko, I. Malysheva, H. Handroos, A. Kovartsev, Faster than real-time simulation of mobile crane dynamics using digital twin concept, J. Phys. Conf. Ser. 1096 (1) (2018), 012071, https://doi.org/10.1088/1742-6596/1096/1/012071.
[21] T. Moi, A. Cibicik, T. Rølvåg, Digital twin based condition monitoring of a knuckle boom crane: an experimental study, Eng. Fail. Anal. 112 (2020), 104517, https://doi.org/10.1016/j.engfailanal.2020.104517.
[22] Y.S. Duarte, J. Szpytko, A digital twins concept model for integrated maintenance: a case study for crane operation, J. Intell. Manuf. 32 (7) (2020) 1863–1881, https://doi.org/10.1007/s10845-020-01689-5.
[23] S.K. Kang, R. Jin, X. Deng, R.S. Kenett, Challenges of modeling and analysis in cyber manufacturing: a review from a machine learning and computation perspective, J. Intell. Manuf. (2020) 1–14, https://doi.org/10.1007/s10845-021-01817-9.
[24] H. Kim, C.R. Ahn, D. Engelhaupt, S. Lee, Application of dynamic time warping to the recognition of mixed equipment activities in cycle time measurement, Autom. Constr. 87 (2018) 225–234, https://doi.org/10.1016/j.autcon.2017.12.014.

[25] C. Chen, Z. Zhu, A. Hammad, Automated excavators activity recognition and productivity analysis from construction site surveillance videos, Autom. Constr. 110 (2020), 103045, https://doi.org/10.1016/j.autcon.2019.103045.
[26] S. Sepasgozar, Digital twin and web-based virtual gaming technologies for online education: a case of construction management and engineering, Appl. Sci. 10 (13) (2020) 4678, https://doi.org/10.3390/app10134678.
[27] S.A. Deryabin, I.O. Temkin, S.V. Zykov, About some issues of developing digital twins for the intelligent process control in quarries, Proc. Comp. Sci. 176 (2020) 3210–3216, https://doi.org/10.1016/j.procs.2020.09.128.
[28] M.J. Skibniewski, Research trends in information technology applications in construction safety engineering and management, Front. Eng. Manag. 1 (3) (2014) 246–259, https://doi.org/10.15302/J-FEM-2014034.
[29] A. Chan, M.A. Xiaozhi, Y.I. Wen, Z. Xin, F. Xiong, Critical review of studies on building information modeling (BIM) in project management, Front. Eng. Manage. 5 (3) (2018) 394–406, https://doi.org/10.15302/J-FEM-2018203.
[30] D. Zhong, H. Lv, J. Han, Q. Wei, A practical application combining wireless sensor networks and internet of things: safety management system for tower crane groups, Sensors. 14 (8) (2014) 13794–13814, https://doi.org/10.3390/s140813794.
[31] S. Hwang, Ultra-wide band technology experiments for real-time prevention of tower crane collisions, Autom. Constr. 22 (2012) 545–553, https://doi.org/10.1016/j.autcon.2011.11.015.
[32] Z. Yang, Y. Yuan, M. Zhang, X. Zhao, B. Tian, Safety distance identification for crane drivers based on mask R-CNN, Sensors. 19 (12) (2019) 2789, https://doi.org/10.3390/s19122789.
[33] J. Yang, P. Vela, J. Teizer, Z. Shi, Vision-based tower crane tracking for understanding construction activity, J. Comput. Civ. Eng. 28 (1) (2014) 103–112, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000242.
[34] X.P. Li, L.C. Gu, J. Jia, Anti-collision method of tower crane via ultrasonic multi-sensor fusion, in: 2012 Automatic Control and Artificial Intelligence (ACAI), 2012, pp. 522–525, https://doi.org/10.1049/cp.2012.1031.
[35] C. Tao, J. Teizer, Modeling tower crane operator visibility to minimize the risk of limited situational awareness, J. Comput. Civ. Eng. 28 (3) (2012) 04014004, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000282.
[36] L. Gu, X. Kou, J. Jia, Distance measurement for tower crane obstacle based on multi-ultrasonic sensors, in: 2012 IEEE International Conference on Automation Science and Engineering (CASE), 2012, pp. 1028–1032, https://doi.org/10.1109/CoASE.2012.6386344.
[37] M.A. Nadar, C.A. Awakian, H.K. Khoury, An intelligent system for monitoring tower cranes on construction sites, in: ISARC. Proceedings of the International Symposium on Automation and Robotics in Construction 30, 2013, p. 1, https://doi.org/10.22260/ISARC2013/0139.
[38] Y. Li, C. Liu, Integrating field data and 3D simulation for tower crane activity monitoring and alarming, Autom. Constr. 27 (2012) 111–119, https://doi.org/10.1016/j.autcon.2012.05.003.
[39] K. Lalik, I. Dominik, P. Wikaa, J. KwaNiewski, Integrated stress measurement system in tower crane mast, Measurement. 102 (2017) 47–56, https://doi.org/10.1016/j.measurement.2017.01.041.
[40] M.A. Sayed, M.R. Kaloop, D. Kim, E. Kim, Movement identification model of port container crane based on structural health monitoring system, Struct. Eng. Mech. 50 (1) (2014) 105–119, https://doi.org/10.12989/sem.2014.50.1.105.
[41] Y. Fang, J. Chen, Y.K. Cho, K. Kim, S. Zhang, E. Perez, Vision-based load sway monitoring to improve crane safety in blind lifts, J. Struct. Integ. Mainten. 3 (4) (2018) 233–242, https://doi.org/10.1080/24705314.2018.1531348.
[42] A. Shapira, S. Filin, A. Wicnudel, Quantitative analysis of blind tower-crane lifts using laser-scanning information, Constr. Innov. 14 (3) (2014) 383–403, https://doi.org/10.1108/CI-09-2013-0039.
[43] G. Lee, J. Cho, S. Ham, T. Lee, G. Lee, S.H. Yun, H.J. Yang, A BIM-and sensor-based tower crane navigation system for blind lifts, Autom. Constr. 26 (2012) 1–10, https://doi.org/10.1016/j.autcon.2012.05.002.
[44] Y. Fang, Y.K. Cho, F. Druso, J. Seo, Assessment of operator's situation awareness for smart operation of mobile cranes, Autom. Constr. 85 (2017) 65–75, https://doi.org/10.1016/j.autcon.2017.10.007.
[45] G. Lee, H.H. Kim, C.J. Lee, S.I. Ham, S.H. Yun, H. Cho, A laser-technology-based lifting-path tracking system for a robotic tower crane, Autom. Constr. 18 (7) (2009) 865–874, https://doi.org/10.1016/j.autcon.2009.03.011.
[46] J. Li, H. Li, W. Umer, H. Wang, X. Xing, S. Zhao, J. Hou, Identification and classification of construction equipment operators' mental fatigue using wearable eye-tracking technology, Autom. Constr. 109 (2020), 103000, https://doi.org/10.1016/j.autcon.2019.103000.
[47] I.M. Rezazadeh, X. Wang, M. Firoozabadi, M.R.H. Golpayegani, Using affective human-machine interface to increase the operation performance in virtual construction crane training system: a novel approach, Autom. Constr. 20 (3) (2011) 289–298, https://doi.org/10.1016/j.autcon.2010.10.005.
[48] H. Li, G. Chan, M. Skitmore, Multiuser virtual safety training system for tower crane dismantlement, J. Comput. Civ. Eng. 26 (5) (2012) 638–647, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000170.
[49] P. Marshall, A. Hirmas, M. Singer, Heinrich's pyramid and occupational safety: a statistical validation methodology, Saf. Sci. 101 (2018) 180–189, https://doi.org/10.1016/j.ssci.2017.09.005.
[50] F. Rauscher, O. Sawodny, Modeling and control of tower cranes with elastic structure, IEEE Trans. Control Syst. Technol. 29 (1) (2020) 64–79, https://doi.org/10.1109/TCST.2019.2961639.
[51] S. Wang, Y. Zhang, Y. Tian, M. Dong, Research on tower crane structure damage detection based on AR model, in: 2017 IEEE 2nd Advanced Information Technology, Electronic and Automation Control Conference (IAEAC), 2017, pp. 2484–2488, https://doi.org/10.1109/IAEAC.2017.805447.
[52] H. Kenan, O. Azelolu, Design of scaled down model of a tower crane mast by using similitude theory, Eng. Struct. 220 (2020) 110985, https://doi.org/10.1016/j.engstruct.2020.110985.
[53] R. Čápková, A. Kozáková, Experimental modelling and control of a tower crane, in: 2019 22nd International Conference on Process Control (PC19), 2019, pp. 97–101, https://doi.org/10.1109/PC.2019.8815327.
[54] M. Zhang, Y. Zhang, B. Ji, C. Ma, X. Cheng, Modeling and energy-based sway reduction control for tower crane systems with double-pendulum and spherical-pendulum effects, Measure. Control. 53 (1–2) (2020) 141–150, https://doi.org/10.1177/0020294019877492.
[55] Y. Ushio, T. Saruwatari, Y. Nagano, Elastoplastic FEM analysis of earthquake response for the field-bolt joints of a tower-crane mast, Adv. Comp. Design. 4 (1) (2019) 53–72, https://doi.org/10.12989/acd.2019.4.1.053.
[56] W. Chen, X. Qin, Z. Yang, P. Zhan, Wind-induced tower crane vibration and safety evaluation, J. Low Frequency Noise, Vibrat. Active Control. 39 (2) (2020) 297–312, https://doi.org/10.1177/1461348419847306.
[57] D. Voisin, G. Grillaud, C. Solliec, A. Beley-Sayettat, J.L. Berlaud, A. Miton, Wind tunnel test method to study out-of-service tower crane behaviour in storm winds, J. Wind Eng. Ind. Aerodyn. 92 (7–8) (2014) 687–697, https://doi.org/10.1016/j.jweia.2004.03.005.
[58] N.D. Zrnic, S.M. Bosnjak, V.M. Gasic, M.A. Arsic, Z.D. Petkovic, Failure analysis of the tower crane counter-jib, Proc. Eng. 10 (2011) 2238–2243, https://doi.org/10.1016/j.proeng.2011.04.370.
[59] J.U. Cho, S.H. Kim, Analysis on the structural safety of the tower crane under load, J. Korea Acad. Indust. Cooperat. Soc. 11 (10) (2010) 3641–3646, https://doi.org/10.5762/KAIS.2010.11.10.3614.
[60] M.F. Milazzo, G. Ancione, G. Consolo, Human factors modelling approach: application to a safety device supporting crane operations in major hazard industries, Sustainability. 13 (4) (2021) 2304, https://doi.org/10.3390/su13042304.
[61] C.P. Coutinho, A.J. Baptista, J.D. Rodrigues, Reduced scale models based on similitude theory: a review up to 2015, Eng. Struct. 119 (2016) 81–94, https://doi.org/10.1016/j.engstruct.2016.04.016.
[62] L. Tian, X. Zhang, X. Fu, Collapse simulations of communication tower subjected to wind loads using dynamic explicit method, J. Perform. Constr. Facil. 34 (3) (2020) 04020024, https://doi.org/10.1061/(ASCE)CF.1943-5509.0001434.
[63] D. Vamvatsikos, C.A. Cornell, Applied incremental dynamic analysis, Earthquake Spectra 20 (2) (2004) 523–553, https://doi.org/10.1193/1.1737737.